

OmniCare AI: Executive Pitch Deck

Qualcomm Snapdragon® AI Lab Build & Present Challenge 2026 | Target: HP OmniBook X / HP EliteBook Ultra
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1. The Rural Healthcare Specialist Crisis

Over 70% of India lives rurally, but 80% of specialists practice in major cities. Primary Health Centers (PHCs) lack on-site dermatologists, cardiologists, and pulmonologists. Cloud medical AI fails due to zero cellular signal, high costs, and privacy violations under India's DPDP Act 2023. OmniCare AI turns Snapdragon-powered HP PCs into autonomous, 100% offline diagnostic workstations.

2. 10 Major Edge Advancements Delivered Live (25 Tests Verified)

#	Advancement	Hardware Precision & Clinical Breakthrough
1	Contactless rPPG Camera Vitals	HP True Vision 5MP / POS-Net INT8. Facial chromaticity extraction of HR, SpO2, RR in 8.2ms.
2	12-Lead Paper ECG Digitizer	PTB-XL INT8. Optical grid filter (98.4%) & STEMI/AFib detection in 6.8ms.
3	NEWS2 Score & Shock Index	Royal College of Physicians 7-parameter clinical deterioration & Sepsis-3 score.
4	PMBJP Jan Aushadhi Drug Engine	NLEM 2022 database saves rural patients 82.9% + CYP450 interaction checks.
5	Council of AI Specialists	Multi-Agent Llama-3.2 INT4. 4 parallel sub-agents + CMO consensus arbitration.
6	Handheld POCUS Ultrasound AI	USB-C probe cardiac LVEF % and lung sliding (Seashore vs Barcode sign).
7	Multilingual Speech Synthesizer	Audio discharge counseling in 8 Indian regional languages (Tamil, Hindi, etc.).
8	DICOM 3.0 Web-PACS Micro-Server	Part 10 Secondary Capture with HU windowing & HP Wolf Enclave signing.
9	Differential Privacy Federated Learning	DP-SGD ($\epsilon=1.2$) local gradient noise injection with 0% raw data leaks.
10	HP Smart Sense Dynamic Governor	Thermal-aware profiles: Performance (45 TOPS), Balanced, Eco (26h battery).

3. Snapdragon X Elite & HP Synergies

- **45 TOPS Hexagon NPU:** 100% offload for vision, acoustics, and LLM via QNN Execution Provider (HTP v73) at 4.5W.
- **HP Poly Studio Audio:** AI beamforming isolates faint lung sounds from loud field crowds.
- **HP Wolf Security:** AES-256 GCM hardware-isolated vault with tamper-evident audit chains.
- **26+ Hour Battery:** 2 full days of off-grid screening without needing electricity.
- **HP AI Companion:** Natural-language assistant queries local encrypted records with zero cloud leak.